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80%	20%	2	2	Mathematics

عدد المحاضرات	مفردات المقرر
(1 محاضرة)	الوحدات – الأبعاد – الطول – المساحة – الحجم – الكتلة – الوزن – العجلة – الكثافة
(1 محاضرة)	المجموعات – المتباينات – الفترات – القيمة المطلقة
(1 محاضرة)	المنطق الرياضي – القضايا – الجمل المفتوحة
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(1 محاضرة)	المتسلسلات – المتتاليات
(2 محاضرة)	المصفوفات – المحددات

المراجع

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Mathematics

Mathematical Fundamentals

(1) Dimensions and Units

Every measurement has **two parts**.

- The first is **a number (n)**
- The second is **a unit (u)**

For example, the **length of an object = 40cm**.

There is a difference between dimensions and units. **A dimension** is a measure of a physical variable (without numerical values), while **a unit** is a way to assign a number or measurement to that dimension.

There are **three primary unit systems** in use today:

1. **International System of Units** (SI, commonly called **metric units**).
2. **English Engineering System of Units** (commonly called **English units**).
3. **British Gravitational System of Units** (BG).

Primary dimensions and units

- There are **seven primary dimensions** (sometimes called **basic**) dimensions are defined as **independent** or **fundamental dimensions**, from which other dimensions can be obtained.
- The primary dimensions are: **mass, length, time, temperature, electric current, amount of light, and amount of matter**.

Primary Dimension	Symbol	SI unit	BG unit	English unit
mass	M or m	kg (kilogram)	slug	lbm (pound-mass)
length	L or l	m (meter)	ft (foot)	ft (foot)
time	T or t	s (second)	s (second)	s (second)
temperature	T (sometimes q)	K (Kelvin)	R (Rankine)	R (Rankine)
electric current	I (sometimes i)	A (ampere)	A (ampere)	A (ampere)
amount of light	C (sometimes I)	c (candela)	c (candela)	c (candela)
amount of matter	n (sometimes μ)	mol (mole)	mol (mole)	mol (mole)

All **other dimensions** can be derived as **combinations of these seven primary dimensions**, and these dimensions are then called **secondary dimensions**.

Some common physical quantities and their dimensions

Quantity	Relation	Unit	Dimension
stress	force/area	$\text{N/m}^2 = \text{Pa}$	$[\text{MT}^{-2}\text{L}^{-1}]$
pressure	force/area	$\text{N/m}^2 = \text{Pa}$	$[\text{MT}^{-2}\text{L}^{-1}]$
density	mass/volume	kg/m^3	$[\text{ML}^{-3}]$
strain	displacement/length	1	[1]
moment (of a force)	distance $\times$ force	Nm	$[\text{ML}^2\text{T}^{-2}]$
impulse	force $\times$ time	Ns	$[\text{MLT}^{-1}]$
linear momentum	mass $\times$ velocity	kg m/s	$[\text{MLT}^{-1}]$
angular momentum	distance $\times$ mass $\times$ velocity	kg m ² /s	$[\text{ML}^2\text{T}^{-1}]$
work	force $\times$ distance	Nm=J	$[\text{ML}^2\text{T}^{-2}]$
energy	work	Nm=J	$[\text{ML}^2\text{T}^{-2}]$
power	work/time	Nm/s=W	$[\text{ML}^2\text{T}^{-3}]$

Most **SI units** are used in scientific research. SI is a coherent system of units. A **coherent system** of units is one in which the units of derived quantities are obtained as **multiples** or **sub-multiples** of **certain basic units**.

Macro and Micro Prefixes

Macro Prefixes	Micro Prefixes
Kilo (K) 10^3	milli (m) 10^{-3}
Mega (M) 10^6	(μ) 10^{-6}
Giga (G) 10^9	nano (n) 10^{-9}
Tera (T) 10^{12}	pico (p) 10^{-12}
Peta (P) 10^{15}	femto (f) 10^{-15}
Exa (E) 10^{18}	atto (a) 10^{-18}
Zetta (Z) 10^{21}	zepto (z) 10^{-21}
Yotta (y) 10^{24}	yocto (y) 10^{-24}

Note: The following are **not used** in the SI system.

- deca 10^1 deci 10^{-1}
- hecta 10^2 centi 10^{-2}

Types of quantities

1- Dimensional constant

The quantities whose **values are constant** and they **possess dimensions**. For example, velocity of light in vacuum, universal gas constant etc.

2- Dimensional variables

The quantities whose **values are variable**, and they **possess dimensions**. For example, area, volume, density etc.

3- Dimensionless constants

The quantities whose **values are constant**, but they **do not possess dimensions**. For example, π , 1, 2, 3, etc.

4- Dimensionless variables

The quantities whose **values are variable**, and they **do not have dimensions**, e.g., angle, strain, specific gravity etc.

(2) Length, area and volume

Length, area and volume, Dimensional measures of **one-dimensional**, **two-dimensional**, and **three-dimensional** geometric objects. All three are magnitudes, that represent the “**size**” of an object.

- The length is **the size of a line segment**.
- The area is **the size of a closed region in a plane**.
- The volume **is the size of a solid**.

Multiplying or dividing quantities with units often occurs in everyday life. To calculate the area of a room (for the purposes of fitting a carpet, for example) you would need to multiply the length of the room by the width of a room. when you multiply **two one dimensional quantities together**, you get a **two-dimensional quantities**.

- $\text{Number} \times \text{Length} = \text{Length}$ (one dimension)
- $\text{Number} \times \text{Area} = \text{Area}$ (two dimensions)
- $\text{Number} \times \text{Volume} = \text{Volume}$ (three dimensions)
- $\text{Length} \times \text{Length} = \text{Area}$ (two dimensions)
- $\text{Length} \times \text{Length} \times \text{Length} = \text{Volume}$ (three dimensions)
- $\text{Length} \times \text{Area} = \text{Volume}$ (three dimensions)

The opposite is also true:

- $\text{Area} \div \text{Length} = \text{Length}$
- $\text{Volume} \div \text{Number} = \text{Volume}$

(3) Mass

- **Mass** is a measure of the quantity of matter in an object at rest relative to the observer
- The greater the mass of an object, the more difficult it is to speed it up, slow it down, or change its direction
- Mass is a **scalar** quantity that has **magnitude** but no direction
- Mass is measured in **kilograms** (kg)
 - Sometimes mass may be given in **grams** (g) but this will need to be **converted to kilograms** when used in calculations
 - $1000 \text{ g} = 1 \text{ kg}$
 - $1 \text{ g} = 0.001 \text{ kg}$
 - To convert **g to kg**, **divide** the mass in g by 1000
 - To convert **kg to g**, **multiply** the mass in g by 1000

(4) Weight and gravity

- **Weight** is a gravitational **force** on an object with mass.
- Since weight is a force, it is a **vector** quantity with both **magnitude** and **direction**.
- Weight is measured in **newtons** (N)
 - Because of weight:
 - Objects stay firmly on the ground.
 - Objects will always fall to the ground.
 - Satellites are kept in orbit.

Gravitational field strength

Gravitational field strength is the force per unit mass acting on an object in a gravitational field.

- **On Earth**, this is equal to **9.8 N/kg**

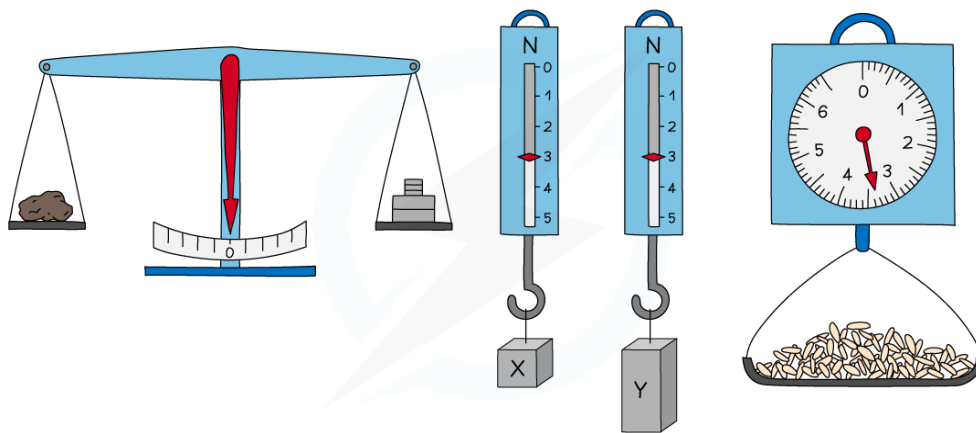
$$g = \frac{W}{m}$$

Where:

- g = gravitational field strength, measured in newtons per kilogram (N/kg).
- W = force of weight, measured in newtons (N).
- m = mass of object, measured in kilograms (kg).

Using a balance

- **The weight** of two objects can be compared using a **balance**
 - Because the gravitational field strength is **constant** everywhere on Earth, this also allows us to measure the **mass** of an object.
- **Balances** can be digital or analogue
 - The object being measured is placed on the balance
 - The reading given is **mass** in kg or g
- **Force meters**, or **newton meters**, consist of a spring and hook
 - The object being measured is hung from the hook
 - The reading given is **weight** in N



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Types of the balances

(5) Density

- Density is **the mass per unit volume of a material**.
- Density **is related to mass and volume** by the following equation:

$$\rho = \frac{m}{V}$$

Where:

- ρ = density, measured in kilograms per meter cubed (kg m^{-3})
- m = mass, measured in kilograms (kg)
- V = volume, measured in meters cubed (m^3)